

SCENARIO 4

A House Full of Generations

SCENARIO AT A GLANCE

Three generations live together under one roof. The grandparents spend most of the day at home, the parents work, and the children attend school and use electronic devices for education and entertainment.

Because the house is active for most of the day and different family members follow different routines, energy consumption is relatively high and difficult to predict.

The family wants to understand when demand is highest and how energy use could be managed more efficiently without reducing anyone's comfort.

What you need to do

- build a realistic historical hourly energy consumption profile for a minimum of 30 consecutive days,
- forecast energy demand for the next 24 hours, with the possibility of extending the forecast to 3 and 7 days,
- use real historical and forecast weather data,
- identify peak demand periods and explain changes in consumption,
- present the results in an application,
- evaluate how renewable energy and better timing of appliance use could reduce grid electricity consumption and household costs.

What you need

You will work with:

- the household profile and description of the residents' behaviour,
- indicative energy consumption parameters for household appliances and systems,
- an example household location for obtaining real historical and forecast weather data,
- your own clearly documented assumptions where additional information is required.

Context

Three generations live together in one detached house.

The grandparents spend much of the day at home, while the parents work and the children attend school. In the afternoon and evening, most family members are usually at home at the same time.

Each generation has different habits and needs. Heating, cooking, laundry, entertainment, computers and household appliances may all contribute to energy demand at different times of the day.

Because the house is rarely completely empty, electricity consumption remains significant throughout the day and may increase sharply when several residents use different appliances at the same time.

The family wants a solution that will help them forecast future energy demand, understand the main sources of peak consumption and identify opportunities to use energy more efficiently without requiring everyone to follow the same routine.

Household Profile

multigenerational household	6 residents
2 grandparents	2 working adults
2 school-age children	detached house
residents present during most hours of the day	different daily schedules
electric heating or a heat pump	relatively high number of household appliances
significant morning and evening activity	

Appliances and Indicative Energy Consumption Parameters

APPLIANCE / SYSTEM	INDICATIVE ENERGY CONSUMPTION
Heat pump / electric heating	2.0–4.0 kW during operation
Refrigerator	0.8–1.2 kWh/day
Freezer	0.6–1.0 kWh/day
Kettle	2.0 kW, typically 3–5 min/use
Coffee machine	1.0–1.5 kW, typically 5–10 min/use
Oven	2.0–2.5 kW during operation
Induction hob	1.5–4.0 kW depending on use
Washing machine	0.6–1.0 kWh/cycle
Tumble dryer	1.5–2.5 kWh/cycle
Dishwasher	0.8–1.2 kWh/cycle
Television	0.08–0.15 kW per unit during operation
Gaming console	0.10–0.20 kW during operation
Computer / laptop	0.05–0.25 kW per device during operation
Wi-Fi router	0.008–0.015 kW, continuous operation
Lighting	0.10–0.30 kW total when active
Phone / tablet charging	0.005–0.02 kWh/device per charge
Standby devices	0.03–0.08 kW combined



EXAMPLE HOUSEHOLD LOCATION

Copenhagen, Denmark

The values provided above are indicative and are intended to serve as input parameters for the challenge. Participants may make reasonable additional assumptions where necessary, provided that these assumptions are clearly documented in the final solution.

Participants should use real historical and forecast weather data for the indicated location.

Challenge

Your task is to create a solution that forecasts the household's energy demand and optimises that demand in terms of cost.

1 Historical Energy Consumption Profile

Based on the provided scenario, prepare a realistic historical electricity consumption profile for the household.

The historical profile should cover **a minimum of 30 consecutive days with hourly resolution**.

The profile should:

- reflect the residents' daily routines,
- include appliance and system operation,
- account for the impact of outdoor temperature,
- include realistic differences between individual days,
- reflect periods of higher and lower household activity,
- reflect the behavioural variability described in the scenario.

The historical profile will serve as input data for the next part of the task.

2 Energy Consumption Forecast

Prepare a model that forecasts the household's hourly energy demand.

The solution must generate a forecast for:

- the next 24 hours.

The solution should also support forecasts for:

- the next 3 days,
- the next 7 days,

depending on data availability.

The forecast should account for predicted weather conditions and the expected household occupancy.

3 Application

Prepare an application that presents the system's results.

The application must display:

- forecast hourly energy consumption,
- forecast total energy consumption for the selected period,
- outdoor temperature forecast,
- hours with the highest expected demand,
- a brief explanation of the expected increase or decrease in consumption.

The user should be able to view a forecast for at least the next 24 hours.

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Renewable Energy Simulator

The solution should include a module for assessing the impact of an investment in renewable energy sources on the household.

The module should allow the user to select the installation capacity in kWp.

For the purpose of the challenge, it may be assumed that the house has suitable conditions for a rooftop photovoltaic installation.

For the selected option, the application should present:

01

estimated annual energy production,

02

**the share of self-generated energy
in covering household demand,**

03

**estimated reduction in electricity
purchased from the grid,**

04

estimated savings,

05

indicative payback period for:

A

a renewable energy installation
without changing current habits,

B

a renewable energy installation
combined with shifting selected
energy-consuming activities
to more favourable periods.

The module should enable comparison of at least several installation capacity options.

We provide common assumptions for electricity tariffs and PV costs for all teams in the shared attachment. For any other assumptions, such as renewable energy production, you are free to choose your own values. Just state them clearly in your solution.

Description of Residents' Behaviour

6:00–9:00

The household becomes very active in the morning.

During this period:

- several people use lighting,
- breakfast is prepared,
- the kettle and coffee machine may be used several times,
- the children prepare for school,
- the grandparents begin their day,
- heating maintains a comfortable indoor temperature.
- the parents prepare for work,

Several appliances may operate at the same time.

9:00–14:00

The parents are usually at work and the children are at school.

The grandparents normally remain at home.

During this period, they may:

- watch television,
- listen to the radio,
- prepare food or drinks,
- use lighting,
- carry out household tasks,
- use the washing machine.

Because somebody is normally at home, heating and other household systems continue to operate throughout the day.

14:00–17:00

The children return from school and household activity increases.

During this period:

- computers and laptops may be used,
- television or gaming devices may operate,
- food and drinks may be prepared,
- electronic devices are charged,
- the grandparents remain active in the house.

One or both parents may also return during this period depending on their schedule.

17:00–22:00

Most or all residents are usually at home.

This is often the period with the highest energy demand.

During this time:

- dinner is prepared,
- the oven or hob may be used,
- the dishwasher may operate,
- the washing machine or tumble dryer may be used,
- televisions, computers and gaming devices are active,
- lighting use increases,
- heating maintains comfortable conditions throughout the house.

Several high-power appliances may operate simultaneously.

22:00–6:00

Household activity gradually decreases.

The following remain active:

- refrigerator and freezer,
- heating system,
- Wi-Fi router,
- standby devices,
- phone and tablet charging.

Behavioural Variability

The household profile is not identical every day.

The following should be taken into account:

- the parents may occasionally work from home,
- the grandparents may leave the house for several hours,
- children's schedules differ between school days and weekends,
- meal times may vary,
- washing and drying are performed only on selected days,
- dishwasher use varies between days,
- visitors or family gatherings may significantly increase household activity,
- weekend occupancy is usually higher than on weekdays,
- heating-related energy consumption changes with weather conditions,
- the entire family may occasionally leave the house for several hours or a full day,
- several appliances may operate at the same time, creating temporary peaks in demand.

Required Outcome

At the end of the challenge, the solution should include:

- | | |
|---|--|
| ☐ 1 a historical hourly energy consumption profile covering a minimum of 30 consecutive days, | ☐ 2 an energy consumption forecasting model, |
| ☐ 3 an hourly forecast for the next 24 hours, | ☐ 4 the ability to present forecasts for the next 3 and 7 days, |
| ☐ 5 an application presenting the results, | ☐ 6 identification of the hours with the highest expected demand, |
| ☐ 7 a brief explanation of changes in forecast energy consumption, | ☐ 8 a simulator showing the impact of a renewable energy investment on the household's energy balance and costs. |

Final Objective

The solution should make it possible to answer the following questions:

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- Q1** What energy demand can be expected in this household over the next 24 hours, 3 days, and 7 days?
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- Q2** During which hours will energy demand be highest?
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- Q3** What changes in household habits could be proposed to optimise the use of renewable energy and electricity from the grid?
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- Q4** How could the selected renewable energy investment affect the amount of electricity purchased from the grid, household costs, and the investment payback period?
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